

Practical 2

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Import pandas library

```
[1]: import pandas as pd
```

```
[2]: import numpy as np
```

```
[3]: import matplotlib.pyplot as plt
```

Read the CSV file (replace 'your_file.csv' with the correct file path)

```
[5]: df=pd.read_csv(r"C:\Users\CNLAB02\Desktop\student_dataset.csv")
```

```
[9]: # Display the first few rows of the DataFrame  
df.head()
```

```
[9]:
```

	stud_id	stud_name	subject1	subject2	subject3	subject4
0	1	vijay	21	14	100	65
1	2	ajay	78	83	46	29
2	3	jay	59	46	29	39
3	4	om	38	37	85	14
4	5	omkar	80	71	33	53

```
[10]: df
```

```
[10]:
```

	stud_id	stud_name	subject1	subject2	subject3	subject4
0	1	vijay	21	14	100	65
1	2	ajay	78	83	46	29
2	3	jay	59	46	29	39
3	4	om	38	37	85	14
4	5	omkar	80	71	33	53
5	6	akash	55	76	42	30
6	7	rishi	31	65	56	47
7	8	rishikesh	58	19	55	81
8	9	jitendra	100	24	54	36
9	10	kundan	54	87	38	16
10	11	tushar	63	21	92	31

11	12	abhay	52	53	90	54
12	13	abhijit	41	75	63	66
13	14	abhinav	16	35	20	33
14	15	adarsh	46	93	68	31
15	16	adesh	33	32	16	18
16	17	adhar	69	18	62	59
17	18	adi	57	58	17	19
18	19	aditya	49	24	59	25
19	20	ajay	59	76	99	36
20	21	ajinkya	81	94	47	22
21	22	ajish	17	21	42	52
22	23	ajit	51	57	41	62
23	24	akansha	87	68	86	81
24	25	akhil	18	36	84	15
25	26	akshat	75	100	47	24
26	27	amit	52	25	88	44
27	28	amar	17	72	10	62
28	29	amrik	72	72	75	62
29	30	anand	58	98	18	83
30	31	anil	12	87	22	18
31	32	ankit	67	92	85	92
32	33	anuj	54	33	53	22
33	34	anuraj	26	73	72	11
34	35	anup	96	86	75	84
35	36	arjun	26	47	57	44
36	37	arnav	71	19	33	85
37	38	aseem	90	82	89	16
38	39	ashwin	70	92	50	71
39	40	ayaz	65	65	70	18
40	41	bali	39	64	70	63
41	42	bharat	89	25	65	63
42	43	bhaves	90	63	79	89
43	44	bhishma	59	83	72	44
44	45	deepak	32	41	96	53
45	46	dikap	21	39	24	43
46	47	dheeraj	41	67	36	100
47	48	dinesh	35	20	68	81
48	49	ganesh	22	11	40	65
49	50	harsh	11	97	17	47

```
[11]: s=pd.notnull(df["subject1"])
      print(s)
```

```
#Checking for missing values using
0      True
1      True
```

2	True
3	True
5	True
6	True
7	True
8	True
9	True
10	True
11	True
12	True
13	True
14	True
15	True
16	True
17	True
18	True
19	True
20	True
21	True
22	True
23	True
24	True
25	True
26	True
27	True
28	True
29	True
30	True
31	True
32	True
33	True
34	True
35	True
36	True
37	True
38	True
39	True
40	True
41	True
42	True
43	True
44	True
45	True
46	True
47	True
48	True
49	True

Name: subject1, dtype: bool

```
[12]: df.isnull() #Checking for missing values using isnull()
```

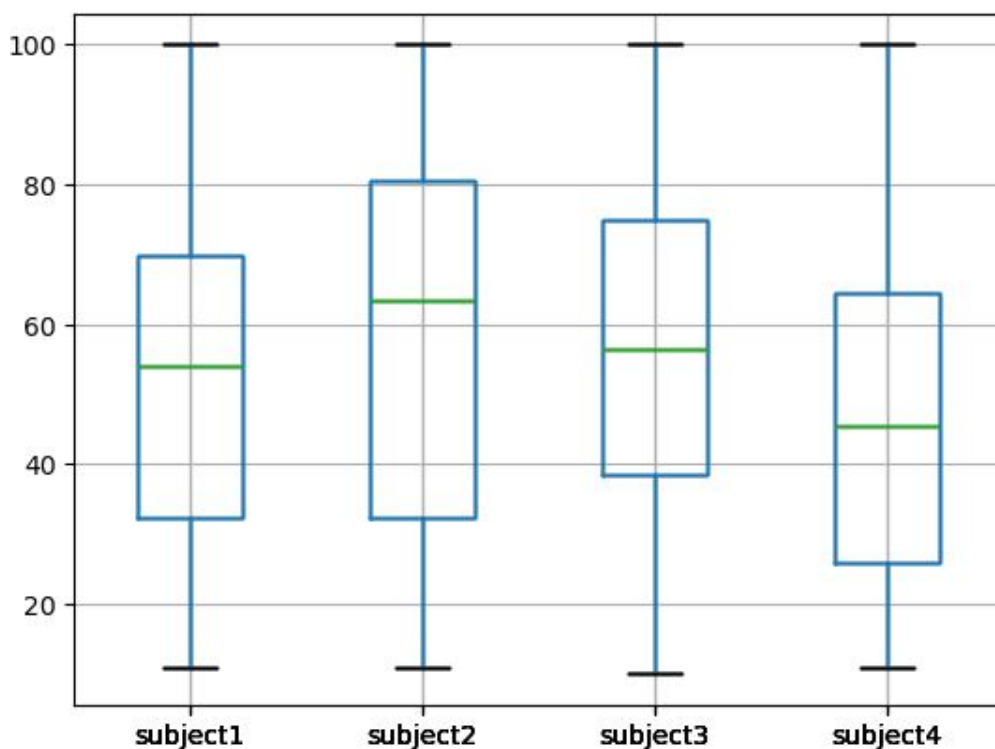
```
[12]:
```

	stud_id	stud_name	subject1	subject2	subject3	subject4
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
5	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	False
8	False	False	False	False	False	False
9	False	False	False	False	False	False
10	False	False	False	False	False	False
11	False	False	False	False	False	False
12	False	False	False	False	False	False
13	False	False	False	False	False	False
14	False	False	False	False	False	False
15	False	False	False	False	False	False
16	False	False	False	False	False	False
17	False	False	False	False	False	False
18	False	False	False	False	False	False
19	False	False	False	False	False	False
20	False	False	False	False	False	False
21	False	False	False	False	False	False
22	False	False	False	False	False	False
23	False	False	False	False	False	False
24	False	False	False	False	False	False
25	False	False	False	False	False	False
26	False	False	False	False	False	False
27	False	False	False	False	False	False
28	False	False	False	False	False	False
29	False	False	False	False	False	False
30	False	False	False	False	False	False
31	False	False	False	False	False	False
32	False	False	False	False	False	False
33	False	False	False	False	False	False
34	False	False	False	False	False	False
35	False	False	False	False	False	False
36	False	False	False	False	False	False
37	False	False	False	False	False	False
38	False	False	False	False	False	False
39	False	False	False	False	False	False
40	False	False	False	False	False	False
41	False	False	False	False	False	False
42	False	False	False	False	False	False
43	False	False	False	False	False	False

44	False	False	False	False	False	False
45	False	False	False	False	False	False
46	False	False	False	False	False	False
47	False	False	False	False	False	False
48	False	False	False	False	False	False
49	False	False	False	False	False	False

```
[13]: col=['subject1','subject2','subject3','subject4']
```

```
[19]: col=['subject1','subject2','subject3','subject4']
df.boxplot(col)
plt.show()
```



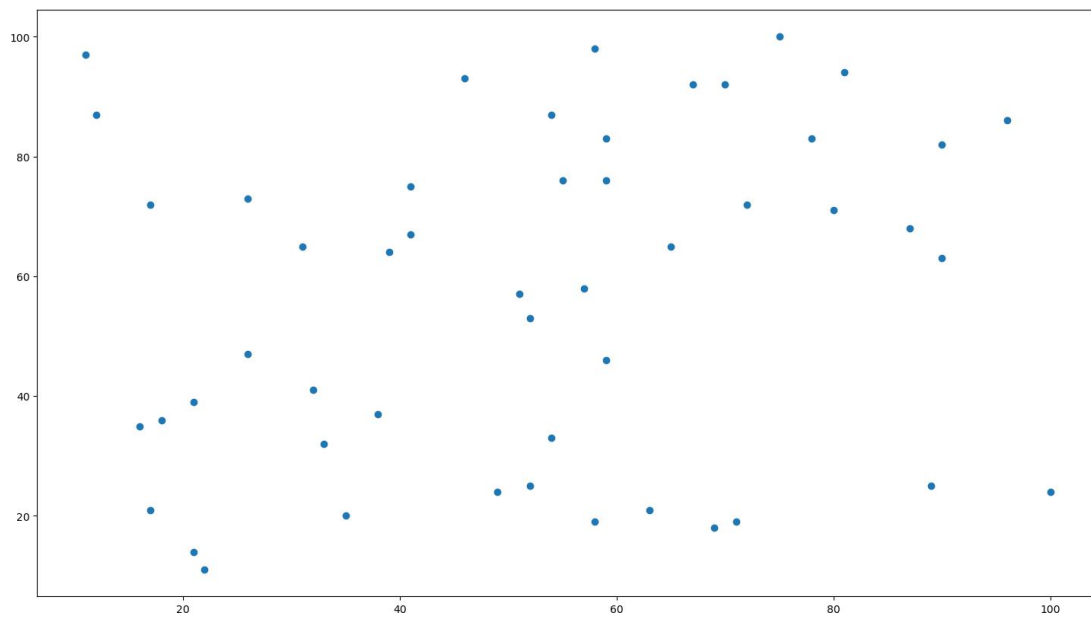
```
[23]: import matplotlib.pyplot as plt

# Create a figure and axes with specified size
fig, ax = plt.subplots(figsize=(18, 10)) # Remove the extra parenthesis

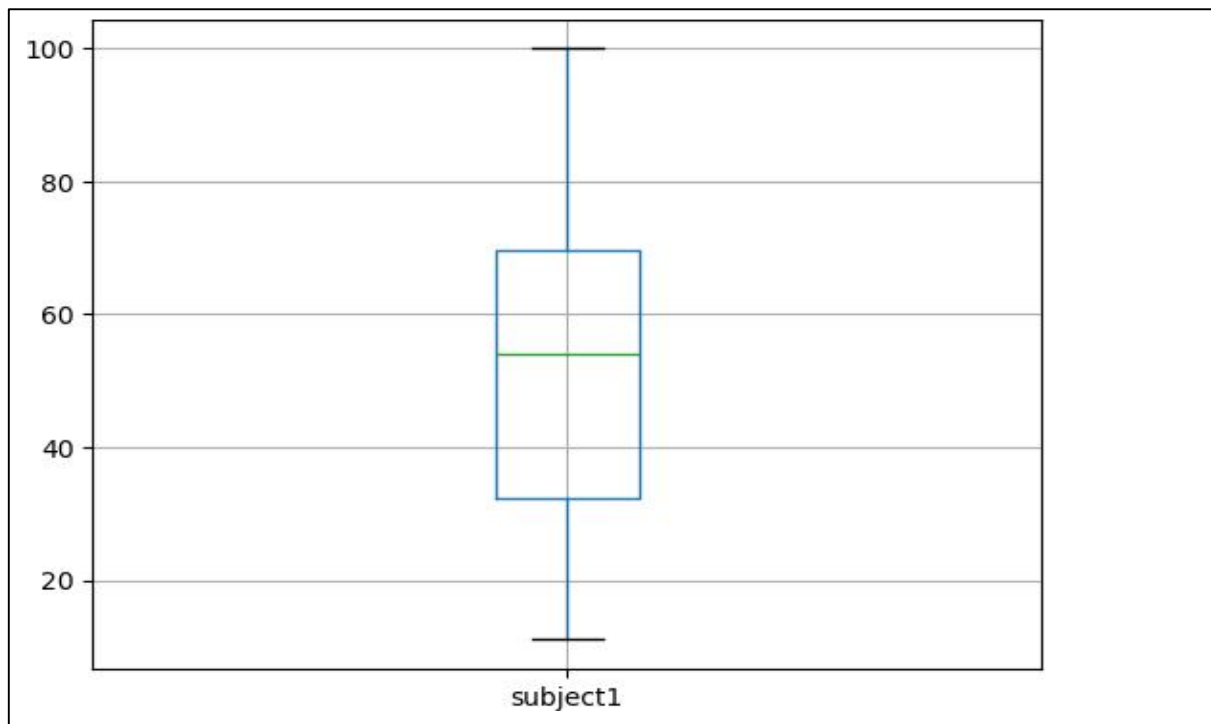
# Create a scatter plot of 'subject1' vs. 'subject2'
ax.scatter(df['subject1'], df['subject2'])

# Display the plot
```

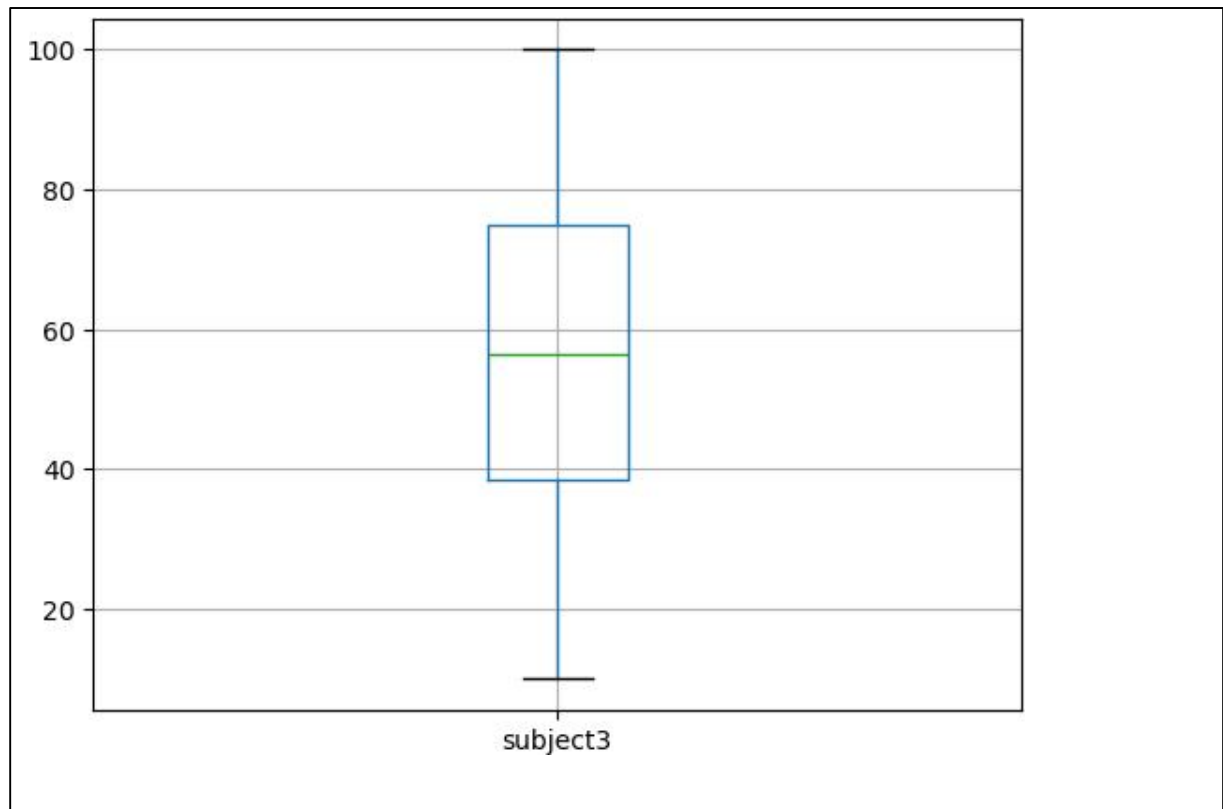
```
plt.show()
```



```
[26] col=['subject1']  
df.boxplot(col)  
plt.show()
```



```
[28] col=['subject3']  
df.boxplot(col)  
plt.show()
```



```
[29]: col2=['subject2']  
df.boxplot(col2)  
plt.show()
```

